

Introduction to Metaoptisc

Foundations for Modern Optics and Metalenses

Ahmad Hasan Sakib

Department of Physics
Shahjalal University of Science and Technology

July 29, 2026

Light as an Electromagnetic Wave

Light is a **transverse electromagnetic (EM) wave** consisting of mutually coupled electric and magnetic fields, $E(r, t)$ and $B(r, t)$, which propagate through free space without requiring a material medium.

Maxwell's Equations in Free Space

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}$$

In free space, there are no free charges ($\rho = 0$) and no free currents ($\mathbf{J} = 0$).

Electromagnetic Wave Equation

Combining Maxwell's equations leads to the wave equations for the electric and magnetic fields:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0$$

$$\nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0$$

Since

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}},$$

the wave equations become

$$\nabla^2 \psi - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = 0, \quad \psi = \mathbf{E}, \mathbf{B}.$$

Descriptive Parameters of a Monochromatic Wave

For a monochromatic wave[Single-frequency, flat and Parallel plane wavefront],

$$E(z, t) = E_0 \cos(kz - \omega t + \phi)$$

- **Amplitude (E_0):** Determines the wave intensity, $I \propto E_0^2$.
- **Phase (ϕ):** Specifies the position within an oscillation cycle; controls the wavefront shape and propagation.
- **Period (T):** Time for one complete oscillation
- **Wavelength (λ):** Distance between successive points of equal phase
- **Propagation Speed (v):** Speed of light propagation

What Does It Mean to Control Light?

- **Neutral-density filters:** Control the **amplitude** (intensity) of light.
- **Dichroic and colored filters:** Control the **wavelength** through selective absorption or reflection.
- **Polarizers and wave plates:** Control the **polarization** state.
- **Diffraction gratings and prisms:** Control the **propagation direction**.
- **Lens:** Primarily control the **phase** of light, thereby shaping the wavefront.

What Does a Lens Actually Do?

- A curved lens has a position-dependent thickness,

$$L = L(x, y).$$

- Different parts of the wave travel different optical path lengths
- The transmitted wavefront becomes curved, causing the light to converge (or diverge).
- **Fundamental Principle:** A lens controls light by engineering the spatial phase profile,

$$\Phi(x, y).$$

- Conventional lenses achieve this indirectly by varying the optical path length,

$$\text{OPL} = n L(x, y).$$

Plane Slab vs. Convex Lens

Plane Slab

Thickness	Constant
Phase	Constant
$\nabla\Phi$	0
Wavefront	Planar
Rays	Undeviated

Convex Lens

Thickness	Varies with (x, y)
Phase	Spatially varying
$\nabla\Phi$	Nonzero
Wavefront	Curved
Rays	Focused

Spatially varying phase $\implies$ Wavefront shaping

Fundamental Idea

To control light using a lens, we must learn to control its **phase**.

Lens Design $\iff$ Phase Engineering

- Conventional lenses control phase by varying the optical path length.

Fundamental Question

*What phase profile $\Phi(x, y)$ must a **flat surface** impose on an incident wave so that the outgoing wavefront focuses correctly?*

Curved Surface $\implies$ Phase Engineering $\implies$ Flat Surface (Metalens)

The geometry is flat; the phase is engineered.

Thought Experiment: A Programmable Flat Surface

Phase Profile

Suppose a flat metasurface imposes a **linearly increasing phase**

$$\phi(x) = kx.$$

The phase increases from left to right.

0° 30° 60° 90° 120°

Left |—————| Right

Think Before Calculating

- What Will happen to the emerging wavefront ?

What Does Phase Mean?

A monochromatic wave can be written as

$$E(x, t) = A \cos(\omega t - \phi).$$

Physical Interpretation

A phase shift is simply a **time delay**.

$$\phi = \omega \Delta t$$

or equivalently

$$\Delta t = \frac{\phi}{\omega}.$$

Phase is time measured in angular units.

Phase Means Delay

Suppose the incident wavefront reaches every point simultaneously.

$$t = 0$$

Every point receives the incoming wave at the same instant.

The metasurface does only one thing:

Impose a position-dependent delay $\Delta t(x) = \frac{\phi(x)}{\omega}$.

Different phase $\iff$ Different delay.

Different Points Fire at Different Times

Phase	Delay	Response
0°	0	Fires first
30°	Small	Slight delay
60°	Larger	Delayed
90°	Larger	More delayed
120°	Maximum	Fires last

Left side is ahead. Right side is delayed.

Huygens' Principle

Each point on the metasurface acts as a secondary source.

- Left side starts oscillating first.
- Its wavelet has more time to expand.
- Right side starts later.
- Its wavelet is therefore smaller.

Wavelet radius $\propto$ time since emission.

Formation of the New Wavefront

The outgoing wavefront is the common tangent to all Huygens wavelets.

Different delays $\implies$ Tilted wavefront

Since light propagates normal to the wavefront,

Tilted wavefront $\implies$ Tilted beam.

Phase Profile $\Rightarrow$ Time Delay

$\Rightarrow$ Different Huygens Wavelets

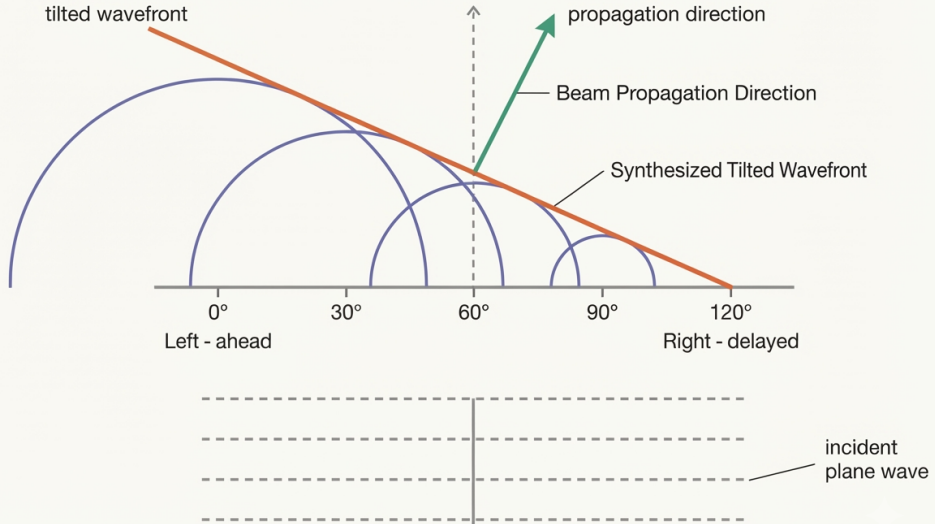
$\Rightarrow$ Tilted Wavefront

$\Rightarrow$ Beam Steering

A metasurface controls light by engineering its phase.

Thought Experiment

Figure 12.4: Wavefront Synthesis and Beam Steering from a Phased Array.



Thought Experiment: A Quadratic Phase Profile

Suppose

A flat metasurface imposes the phase profile

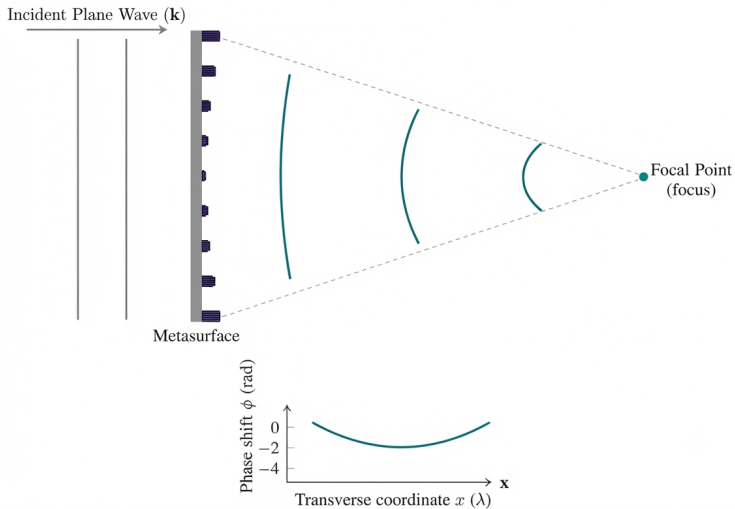
$$\phi(x) = x^2.$$

Question

What happens to the incident plane wavefront after passing through the metasurface?

A Quadratic Phase Profile

Figure 3.4: Schematic of a metalens focusing a normally incident plane wave.



Key Takeaway

Fundamental Idea

A metasurface introduces a **position-dependent phase shift** across its surface. Physically, this is equivalent to introducing a **position-dependent time delay** (retardation) in the transmitted wave.

Spatially Varying Phase $\Longleftrightarrow$ Spatially Varying Time Delay

$$\phi(x, y) \Longleftrightarrow \Delta t(x, y) = \frac{\phi(x, y)}{\omega}$$

Generalized Snell's Law

$$n_t \sin \theta_t - n_i \sin \theta_i = k \frac{d\phi}{dx}$$

Meaning of Each Term

- n_i : Refractive index of the incident medium.
- n_t : Refractive index of the transmitted medium.
- θ_i : Angle of incidence (measured from the surface normal).
- θ_t : Angle of refraction (transmission angle).
- k : wavenumber.
- $\phi(x)$: Phase profile imposed by the metasurface.
- $\frac{d\phi}{dx}$: Spatial phase gradient (phase change per unit distance).

Mathematical Interpretation of a Linear Phase Gradient

The generalized Snell's law is

$$n_t \sin \theta_t - n_i \sin \theta_i = \frac{\lambda_0}{2\pi} \frac{d\phi}{dx}.$$

Suppose the metasurface imposes the phase profile

$$\phi(x) = kx,$$

where k is a constant.

Mathematical Interpretation of a Linear Phase Gradient

Differentiating,

$$\frac{d\phi}{dx} = k.$$

Since the phase gradient is constant, the right-hand side of the generalized Snell's law is also constant,

$$n_t \sin \theta_t - n_i \sin \theta_i = \frac{\lambda_0 k}{2\pi} = \text{constant}.$$

Therefore,

$$\theta_t = \text{constant}.$$

Mathematical Interpretation of a Linear Phase Gradient

Physical Meaning

Every point on the metasurface redirects light by the **same angle**. Consequently, the outgoing rays remain parallel to one another, producing a **tilted plane wavefront** (beam steering) rather than focusing.

$$\phi(x) = kx \implies \frac{d\phi}{dx} = \text{constant} \implies \theta_t = \text{constant} \implies \text{Beam Steering.}$$

The Metalens Phase Profile: The Hyperbolic Lens Law

Problem Statement

A flat metasurface lies in the plane

$$z = 0,$$

and is illuminated by a monochromatic plane wave of vacuum wavelength λ_0 , incident normally ($\theta_i = 0$) from the region $z < 0$. The refractive indices of the incident and transmitted media are n_1 and n_2 , respectively.

The Metalens Phase Profile: The Hyperbolic Lens Law

Objective

Determine the spatial phase profile

$$\Phi(r)$$

that the metasurface must impose so that light transmitted from every radial position r arrives **in phase** at the focal point

$$F = (0, 0, f),$$

thereby producing constructive interference and focusing the incident plane wave.

$$\text{Plane Wave} \implies \text{Metalens} \implies \text{Focus at } F = (0, 0, f)$$

Metalens Hyperbolic Phase Profile

Optical Path Length

A ray transmitted from a point at radial distance r must travel a distance

$$L(r) = \sqrt{r^2 + f^2}$$

to reach the focal point

$$F = (0, 0, f).$$

The central ray ($r = 0$) travels only

$$L(0) = f.$$

The additional optical path traveled by the off-axis ray is

Metalens Hyperbolic Phase Profile

The additional optical path traveled by the off-axis ray is

$$\Delta L(r) = \sqrt{r^2 + f^2} - f.$$

To ensure that all rays arrive at the focus in phase, the metasurface must compensate this path difference by imposing the phase profile

$$\Phi(r) = -\frac{2\pi n_2}{\lambda_0} \left(\sqrt{r^2 + f^2} - f \right)$$

This phase profile is known as the Hyperbolic Phase Profile of a metalens.

From Theory to Reality

Question

So far, we have assumed the existence of a hypothetical surface capable of imposing an arbitrary phase profile,

$$\Phi(x, y).$$

But how can such a surface actually be built?

The next question is therefore

How do we physically realize a metasurface?

More specifically,

- What is the fundamental building block of a metasurface?
- How can a flat surface modify the phase of light?
- How can different phase shifts be programmed at different locations?

The Fundamental Building Block: The Meta-Atom

A metasurface is not a continuous material.

Instead, it is composed of millions of tiny, engineered scattering elements called

Meta-atoms.

Each meta-atom interacts with the incident electromagnetic wave and locally modifies

- phase,
- amplitude,
- polarization,
- or all three simultaneously.

Metasurface = Array of Meta-atoms

The Fundamental Building Block: The Meta-Atom

A metasurface is not a continuous material.

Instead, it is composed of millions of tiny, engineered scattering elements called

Meta-atoms.

Each meta-atom interacts with the incident electromagnetic wave and locally modifies

- phase,
- amplitude,
- polarization,
- or all three simultaneously.

Metasurface = Array of Meta-atoms

What is a Meta-Atom?

Definition

A **meta-atom** is an artificial optical scatterer whose dimensions are smaller than the wavelength of light.

Unlike a natural atom,

- it is engineered rather than naturally occurring,
- its optical response is determined by its geometry,
- not by its electronic energy levels.

Natural Atom $\neq$ Meta-Atom

Natural Atoms vs. Meta-Atoms

Natural Atom	Meta-Atom
Building block of matter	Building block of a metasurface
Size ~ 0.1 nm	Size ~ 100 – 500 nm
Properties fixed by nature	Properties designed by geometry
Response determined by electronic structure	Response determined by shape, size, and material
Cannot be engineered individually	Can be designed independently

What is a Nanopillar?

The most common optical meta-atom is the

Dielectric Nanopillar.

A nanopillar is

- a nanoscale pillar fabricated on a flat substrate,
- typically made from a high-index dielectric material,
- such as TiO_2 , Si, GaN, or SiN.

Typical dimensions are

Diameter $\sim 50 - 300$ nm,
Height $\sim 500 - 1000$ nm,
Period $< \lambda$.

Why Does a Nanopillar Control Phase?

Each nanopillar behaves like a tiny optical resonator (or truncated waveguide).

When light passes through it,

- the electromagnetic field is confined,
- it accumulates a phase delay,
- and then re-radiates into free space.

Changing the pillar geometry changes

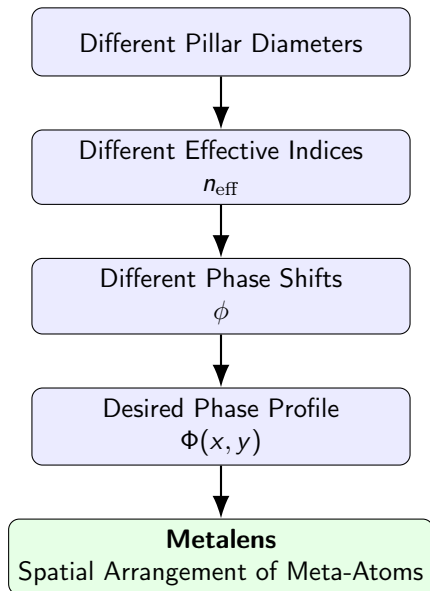
$$n_{\text{eff}}$$

which changes

$$\phi = k_0 n_{\text{eff}} h.$$

Different pillar geometries therefore produce different transmitted phases.

From Nanopillars to a Metalens



- A metasurface is an array of subwavelength **meta-atoms**.
- The most common optical meta-atom is a high-index **dielectric nanopillar**.
- The optical response of a nanopillar is determined by its **geometry** (diameter, height, shape, and material).
- Different nanopillar geometries produce different **effective refractive indices** (n_{eff}).
- Different effective indices introduce different **phase shifts** to the transmitted light.
- By arranging nanopillars with appropriate geometries, an arbitrary spatial phase profile $\Phi(x, y)$ can be realized.
- A **metalens** focuses light by engineering the phase profile rather than the physical thickness of the surface.